

Early-life gut mycobiome core species modulate metabolic health in mice

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The gut microbiome causally contributes to obesity; however, the role of fungi remains understudied. We previously identified three core species of the infant gut mycobiome (*Rhodotorula mucilaginosa*, *Malassezia restricta* and *Candida albicans*) that correlated with body mass index, however their causal contributions to obesity development are unknown. Here we show the effects of early-life colonization by these fungal species on metabolic health in gnotobiotic mice fed standard (SD) or high-fat-high-sucrose (HFHS) diets. Each species resulted in bacterial microbiome compositional and functional differences. *R. mucilaginosa* and *M. restricta* increased adiposity in mice fed SD, while only *R. mucilaginosa* exacerbated metabolic disease. In contrast, *C. albicans* resulted in leanness and resistance to diet-induced obesity. Intestinal nutrient transporter expression was unaffected by the presence of fungi in jejunal enteroids, yet the immune landscape in white adipose tissue was distinctly impacted by each fungal species, suggesting that these phenotypes may be a result of fungal immune regulation. This work revealed that three common fungal colonizers have distinct causal influences on obesity and metabolic inflammation and justifies the consideration of fungi in microbiome research on host metabolism.

The gut microbiome is causally linked to the development of metabolic diseases^{1–3}; however, non-bacterial microbes, including fungi, remain understudied. The early-life period is critical for the development of the gut microbiome in parallel with host systems, and perturbations to this community contribute to the development of several diseases later in life, including obesity^{4–7}. Different risk factors for obesity development that have been attributed to early-life microbiome perturbations, like antibiotics, delivery mode and diet, have also been shown to influence the gut mycobiome^{8–16}. These cross-kingdom perturbations by early-life factors associated with

obesity development suggests the potential involvement of fungi in the etiology of this disease, yet the evidence supporting this is limited.

Studies in animal models have begun to elucidate the involvement of intestinal fungi in energy harvest and host metabolism. *Candida*, one of the most prevalent and abundant genera of the adult gut mycobiome¹⁷, has been studied most extensively. For instance, long-term dietary exposure to *Candida albicans* modulated metabolic hormones and mitigated diet-induced obesity (DIO) in mice¹⁸. In contrast, *Candida parapsilosis* exacerbated obesity development by enhancing dietary lipid harvest via increased lipase activity¹⁹. These

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contrasting outcomes highlight the need to further explore the role of these and other gut mycobiome species in host metabolism and obesity development.

Recent human studies have identified associations between the gut mycobiome and obesity-associated diseases including type 2 diabetes and non-alcoholic fatty liver disease²⁰. For instance, body fat mass, fasting triglycerides, and high-density lipoprotein (HDL) cholesterol were linked to mycobiome clusters in an adult population with or without obesity²¹. In infants, we recently identified altered mycobiome maturation patterns during the first year of life and differential abundance of core fungal species to be correlated with childhood body-mass index (BMI). The abundance of *C. albicans*, *Rhodotorula mucilaginosa* and *Malassezia restricta* at 1 year were associated with BMI scores from 1 to 5 years of age²². While we were able to infer complex relationships between the early-life gut mycobiome and BMI in infants, the causal contributions of these fungal taxa to obesity development remain unknown.

A growing number of human mycobiome studies have revealed associations between gut fungi and chronic inflammatory diseases, including inflammatory bowel disease, asthma and cancer^{23–25}, mainly through fungal interactions with the host immune system. Intestinal fungi are potent modulators of host immunity at both mucosal and distal sites, including the lungs and liver^{26,27}. Our current understanding of bacterial microbiome-derived mechanisms that impact host metabolic health involves alterations in immune cell populations and/or the overall inflammatory tone of white adipose tissue (WAT), the primary site of host energy storage^{28–32}. Given this, we hypothesized that the immunomodulatory properties of intestinal fungi may also influence host energy balance by modulating the WAT immune landscape.

In this work, we used a gnotobiotic approach to explore the causal influence of core early-life gut mycobiome species, *C. albicans*, *R. mucilaginosa* and *M. restricta*, in the development of DIO and metabolic inflammation. We found that early-life colonization with each fungal species impacted bacterial microbiome composition and function in a diet-dependent manner. They also resulted in distinct metabolic phenotypes, with *R. mucilaginosa* and *M. restricta* promoting increased adiposity, although only *R. mucilaginosa* colonization was associated with other markers of metabolic disease. On the other hand, *C. albicans* drove leanness and resistance to DIO, accompanied by broad WAT inflammation. Fungal colonization did not influence metabolite mediated alterations in jejunal enteroid

nutrient transporter expression, supporting an immunomodulatory mechanism underlying these metabolic phenotypes. These findings highlight the involvement of fungi and interkingdom interactions in modulating host metabolism.

Results

Early-life fungal colonizers uniquely modulate microbiome ecology in a model of diet-induced obesity

We developed a gnotobiotic DIO model to interrogate the causal impact of three core human gut mycobiome species on host metabolic health in both male and female mice. Germ-free C57Bl/6J mice were inoculated with the Oligo-Mouse-Microbiota (Oligo-MM12) bacterial consortium³³ alone or in combination with *C. albicans* (B + C), *R. mucilaginosa* (B + R) or *M. restricta* (B + M) twice at the commencement of breeding (Fig. 1A). Dams were kept on a standard mouse diet (SD) during breeding and gestation. While it has been established that the Oligo-MM12 bacterial consortium is heritable³³, whether vertical transmission occurs for the three fungal species in gnotobiotic mice had not been previously determined. Therefore, fungal cultures were also applied to the dams' nipple and abdominal area twice during the first week following birth to promote early-life transfer of these fungal species to the offspring (F1). At 3 weeks of age, F1 mice were then weaned onto a standard (SD) or high-fat-high-sucrose (HFHS) mouse diet until 12 weeks of age to model the development of childhood obesity. This model allowed us to explore the consequences of exposure to individual fungal species during development.

We carried out fungal culture and rDNA detection to determine if each species was able to stably colonize gnotobiotic F1 mice. Independently of sex, both *C. albicans* and *R. mucilaginosa* were able to stably colonize the mice with no difference in fecal fungal growth between the two species when mice consumed either diet. Intriguingly, there was a reduction in viable *C. albicans* and *R. mucilaginosa* in mice fed HFHS (diet $p < 0.0001$; Fig. 1B). *M. restricta* could not be recovered from fecal samples at 6–7 weeks of age, indicating transient early-life exposure. Similarly, amplification of internal transcribed spacer region (ITS) in fecal DNA was only detected in B + C and B + R samples at 12 weeks, with a strong inhibitory effect of HFHS diet on fungal growth in both sexes (diet $p < 0.0001$; Fig. 1C, D). While *C. albicans* is a known pathobiont that can result in disseminated infection²⁶, we did not detect its growth in the blood, liver, or kidneys on either diet, indicating that fungal growth was restricted to the gut.

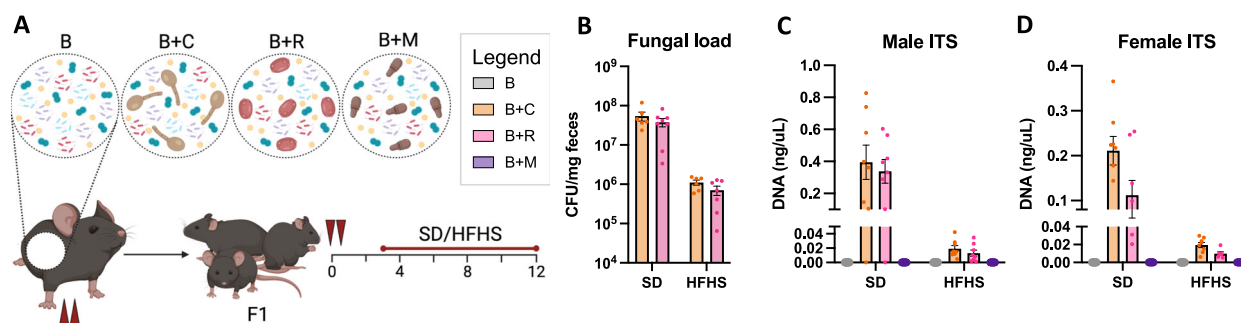


Fig. 1 | Early-life fungal colonization and diet-induced obesity model.

A Gnotobiotic colonization and diet-induced obesity model. Germ-free dams were gavaged twice with the Oligo-MM12 consortium alone (B; gray) or in combination with *C. albicans* (B + C; orange), *R. mucilaginosa* (B + R; pink) or *M. restricta* (B + M; purple); Males—SD: $n_B = 23$, $n_{B+C} = 8$, $n_{B+R} = 10$, $n_{B+M} = 12$; Females—SD: $n_B = 18$, $n_{B+C} = 8$, $n_{B+R} = 8$, $n_{B+M} = 12$; Males—HFHS: $n_B = 22$, $n_{B+C} = 10$, $n_{B+R} = 8$, $n_{B+M} = 12$; Females—HFHS: $n_B = 20$, $n_{B+C} = 12$, $n_{B+R} = 7$, $n_{B+M} = 10$. Twice in the first week of life, offspring (F1) were exposed to the fungal species corresponding to their colonization group. F1 were weaned onto standard diet (SD) or high-fat-high-sucrose diet

(HFHS) until 12 weeks of age. Graphics from BioRender.com. <https://BioRender.com/y96j811>. **B** Fungal colony count in fecal samples at 6–7 weeks of age. One mouse sampled per cage (SD: $n_{B+C} = 6$, $n_{B+R} = 8$; HFHS: $n_{B+C} = 6$, $n_{B+R} = 7$). Two-way ANOVA with Sidak's multiple comparisons test. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. **C, D** Fecal ITS DNA at 12 weeks in males and females, respectively (Male—SD: $n_B = 13$, $n_{B+C} = 7$, $n_{B+R} = 7$, $n_{B+M} = 8$; Male—HFHS: $n_B = 16$, $n_{B+C} = 8$, $n_{B+R} = 8$, $n_{B+M} = 8$; Female—SD: $n_B = 14$, $n_{B+C} = 8$, $n_{B+R} = 8$, $n_{B+M} = 8$; Female—HFHS: $n_B = 14$, $n_{B+C} = 8$, $n_{B+R} = 6$, $n_{B+M} = 8$). Data represented as mean $\pm$ SEM; n indicates number of mice sampled.

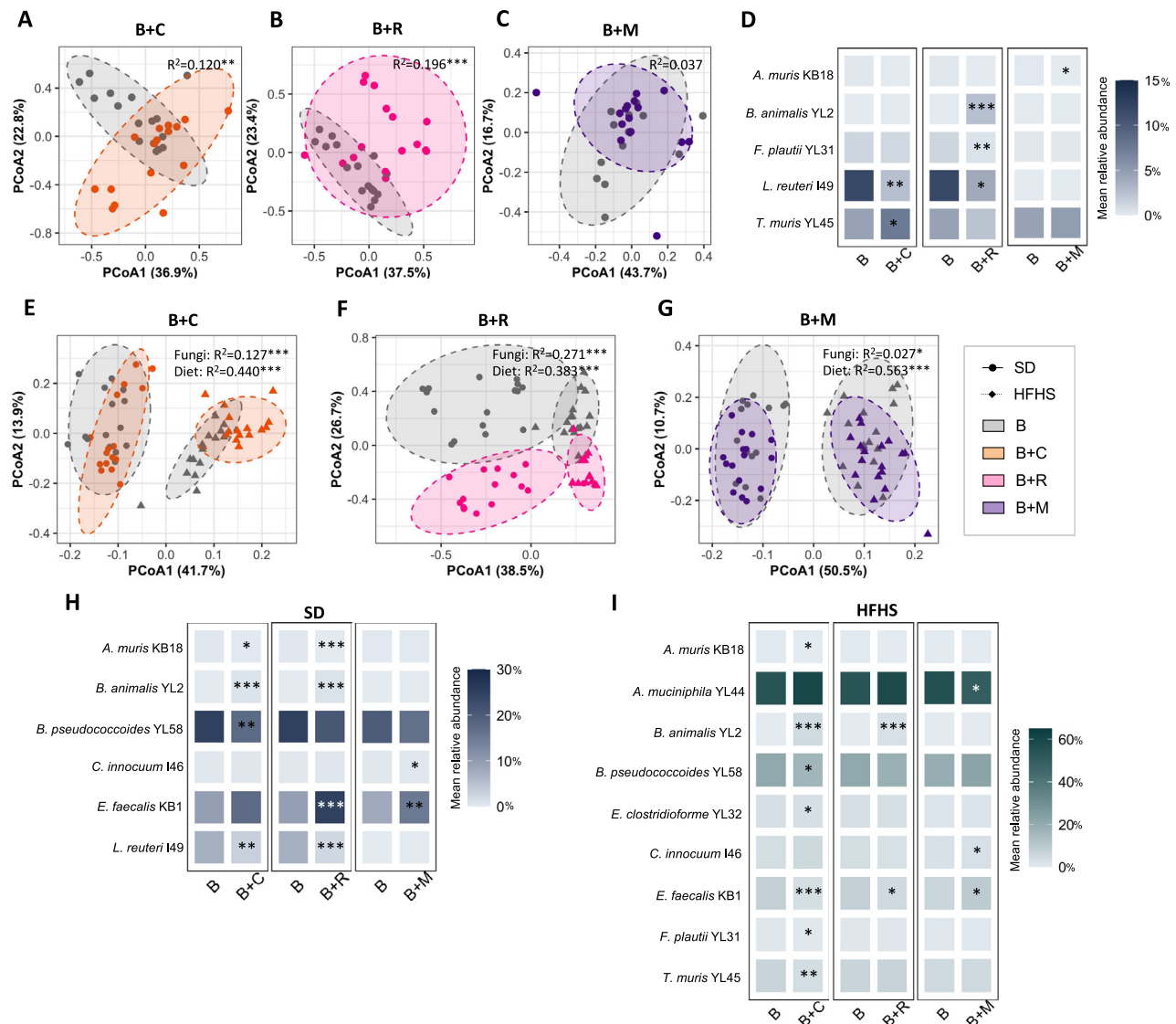


Fig. 2 | Early-life fungal colonization modulates bacterial community composition. **A–C** Bacterial beta-diversity at 3 weeks based on Bray–Curtis dissimilarity index assessed by permutational ANOVA between bacteria only and B + C ($n_B = 16$, $n_{B+C} = 16$), B + R ($n_B = 16$, $n_{B+R} = 16$) and B + M ($n_B = 13$, $n_{B+M} = 16$), respectively. **D** Bacterial relative abundance at 3 weeks between bacteria only and B + C ($n_B = 16$, $n_{B+C} = 16$), B + R ($n_B = 16$, $n_{B+R} = 16$) and B + M ($n_B = 13$, $n_{B+M} = 16$), respectively. **E–G** Bacterial beta-diversity at 12 weeks between bacteria only and B + C (SD: $n_B = 16$, $n_{B+C} = 16$; HFHS: $n_B = 16$, $n_{B+C} = 16$), B + R (SD: $n_B = 16$, $n_{B+R} = 16$; HFHS: $n_B = 16$, $n_{B+R} = 14$) and B + M (SD: $n_B = 14$, $n_{B+M} = 16$; HFHS: $n_B = 16$, $n_{B+M} = 17$), respectively. Fungi and diet effect based on Bray–Curtis Dissimilarity Index

assessed by permutational ANOVA. **H, I** Bacterial relative abundance at 12 weeks between SD-fed or HFHS-fed bacteria only and B + C (SD: $n_B = 16$, $n_{B+C} = 16$; HFHS: $n_B = 16$, $n_{B+C} = 16$), B + R (SD: $n_B = 16$, $n_{B+R} = 16$; HFHS: $n_B = 16$, $n_{B+R} = 14$) and B + M (SD: $n_B = 14$, $n_{B+M} = 16$; HFHS: $n_B = 16$, $n_{B+M} = 17$), respectively. **A–C, E–G** Ellipses represent 95% CI. Dots indicate SD and triangles indicate HFHS. **D, H, I** Data represented as mean relative abundance. Normality assessed by Shapiro–Wilk test and differences calculated by unpaired *T*-test (two-sided) or Mann–Whitney test (two-sided) accordingly. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Source data are provided as a source data file; *n* indicates number of mice sampled. SD standard diet, HFHS high-fat-high-sucrose diet.

These data show that early-life exposure to *C. albicans* and *R. mucilaginosa* resulted in long-term establishment in the gut of gnotobiotic mice, while this only occurred transiently for *M. restricta*.

We then explored the effect of early-life exposure to fungi on the bacterial community. At weaning, before the dietary challenge, we found that exposure to an individual fungal species altered the bacterial community composition early in life. Specifically, both *C. albicans* and *R. mucilaginosa* induced significant shifts in beta-diversity, as measured by the Bray–Curtis Dissimilarity Index, relative to mice exposed to bacteria (B) only (Fig. 2A–C and Table S1). These effects were independent of sex (Table S1). While fungal colonization did not result in changes to alpha-diversity, as measured by the Shannon index (Fig. S1A), it did result in several shifts of bacterial relative abundance. Specifically, *C. albicans* colonization was associated with an increased

relative abundance of *Turicimonas muris* YL45 and decreased *Limosilactobacillus reuteri* I49, whereas *R. mucilaginosa* was associated with increased *Bifidobacterium animalis* YL2 and decreased *Flavonifractor plautii* YL31 and *L. reuteri* I49 (Figs. 2D and S1B). Despite the absence of statistical significance in beta-diversity, *M. restricta* was associated with decreased relative abundance of *Acetivibacter muris* KB18 (Figs. 2D and S1B). These findings indicate that early-life fungal exposure has cross-kingdom consequences for microbiome establishment in a species-specific manner.

We also evaluated the bacterial gut microbiome at the conclusion of the diet challenge at 12 weeks of age. Aligned with previous studies^{34–36}, diet strongly influenced bacterial beta-diversity (Fig. 2E–G), as well as exposure to all three fungi during early life. Again, the strongest effects were induced by *C. albicans* and *R. mucilaginosa*

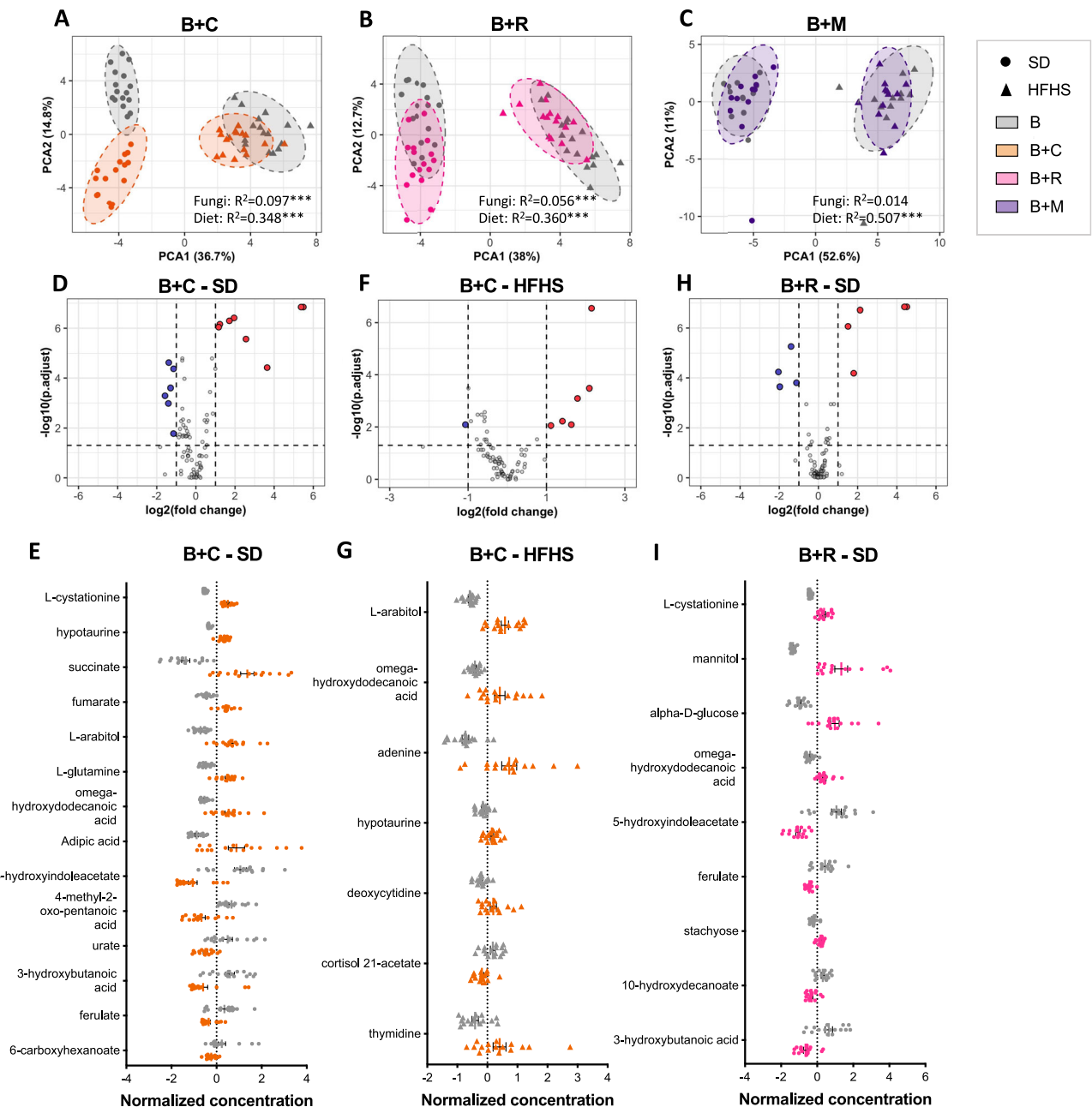


Fig. 3 | Fungal colonization modulates the fecal metabolome and microbiome metabolic function. **A–C** Fecal metabolome in 12-week fecal samples between B and B + C (SD: $n_B = 16$, $n_{B+C} = 16$; HFHS: $n_B = 16$, $n_{B+C} = 16$), B + R (SD: $n_B = 16$, $n_{B+R} = 16$; HFHS: $n_B = 16$, $n_{B+R} = 14$) and B + M (SD: $n_B = 14$, $n_{B+M} = 16$; HFHS: $n_B = 16$, $n_{B+M} = 17$), respectively. Fungi and diet effect based on Euclidean distance and assessed by permutational ANOVA. Ellipses represent 95% CI. Dots indicate SD and triangles indicate HFHS. **D** Fold change analysis of fecal metabolome between B and B + C mice fed SD diet ($n_B = 16$, $n_{B+C} = 16$). **E** Normalized concentrations of metabolites with significant fold change in (**D**) between B and B + C mice fed SD diet ($n_B = 16$, $n_{B+C} = 16$). **F** Fold change analysis of fecal metabolome between B and B + C mice fed HFHS diet ($n_B = 16$, $n_{B+C} = 16$). **G** Normalized concentrations of metabolites with significant fold change in (**F**) between B and B + C mice fed HFHS diet ($n_B = 16$,

$n_{B+C} = 16$). **H** Fold change analysis of fecal metabolome between B and B + R mice fed SD diet ($n_B = 16$, $n_{B+R} = 16$). **I** Normalized concentrations of metabolites with significant fold change in (**H**) between B and B + R mice fed SD diet ($n_B = 16$, $n_{B+C} = 16$). **D**, **F**, **H** Each dot represents an individual metabolite. Highlighted metabolites identified by non-parametric t -tests (two-sided) have $p.adjust$ (false discover rate adjustment) < 0.05 denoted by the dashed horizontal line and fold change ≥ 2 or ≤ 0.5 denoted by the dashed vertical lines. **E**, **G**, **I** Concentrations underwent median normalization, square root transformation and Pareto scaling. Each dot represents an individual mouse. **A–C**, **E**, **G**, **I** B = gray; B + C = orange; B + R = pink; B + M = purple. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$. Source data are provided as a source data file; n indicates number of mice sampled. SD standard diet, HFHS high-fat-high-sucrose diet.

(Fig. 2E–G and Table S2). A sex effect was detected in B + C mice and approached significance in B + R and B + M mice (Table S2). On SD, no alterations in alpha-diversity were observed (Fig. S1C), though all three fungal species influenced the relative abundance of several bacterial strains. *C. albicans* colonization resulted in an increased abundance of *B. animalis* YL2, and decreased *A. muris* KB18, *Blautia pseudococcoides*

YL58 and *L. reuteri* I49 (Figs. 2H and S1D). Similarly, *R. mucilaginosa* was associated with an increased abundance of *B. animalis* YL2 and *Enterococcus faecalis* KB1, and decreased *A. muris* KB18 and *L. reuteri* I49, and (Figs. 2H and S1D). Meanwhile, *M. restricta* was associated with decreased abundance of *Clostridium innocuum* I46 and increased *E. faecalis* KB1 (Figs. 2H and S1D).

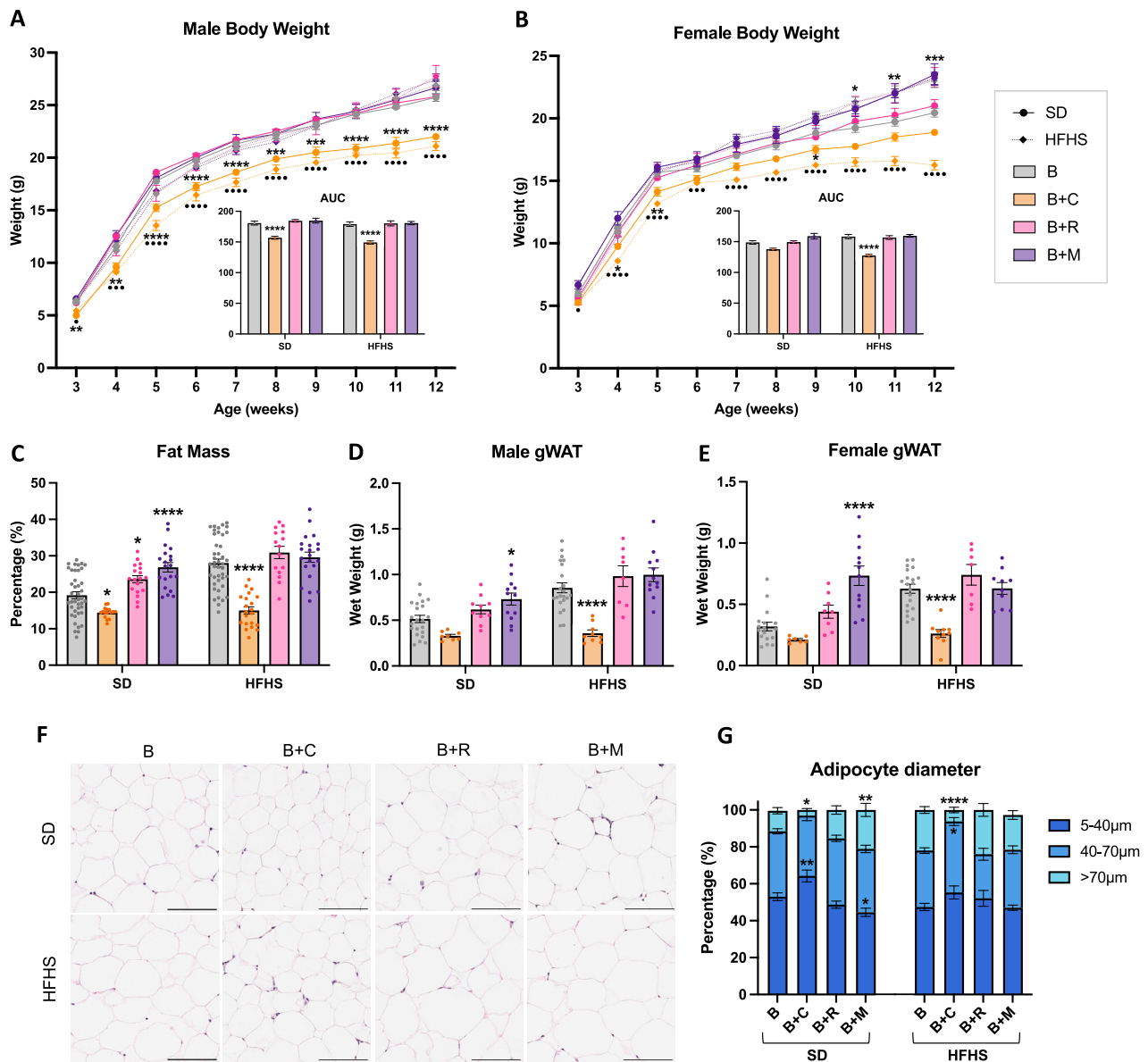


Fig. 4 | Gut fungi differentially modulate obesity development. **A, B** Body weight from 3–12 weeks with trapezoidal area under the curve (AUC) in males and females, respectively (Male–SD: $n_B = 23$, $n_{B+C} = 8$, $n_{B+R} = 10$, $n_{B+M} = 12$; Male–HFHS: $n_B = 22$, $n_{B+C} = 9$, $n_{B+R} = 8$, $n_{B+M} = 12$; Female–SD: $n_B = 18$, $n_{B+C} = 8$, $n_{B+R} = 8$, $n_{B+M} = 12$; Female–HFHS: $n_B = 20$, $n_{B+C} = 12$, $n_{B+R} = 7$, $n_{B+M} = 10$). Significant differences indicated by asterisk for SD and dots for HFHS. **C** Fat mass percentage independent of sex (SD: $n_B = 41$, $n_{B+C} = 16$, $n_{B+R} = 18$, $n_{B+M} = 20$; HFHS: $n_B = 42$, $n_{B+C} = 21$, $n_{B+R} = 15$, $n_{B+M} = 21$). **D, E** WAT wet weight in males and females, respectively (Male–SD: $n_B = 23$, $n_{B+C} = 8$, $n_{B+R} = 10$, $n_{B+M} = 12$; Male–HFHS: $n_B = 22$, $n_{B+C} = 9$, $n_{B+R} = 8$, $n_{B+M} = 12$; Female–SD: $n_B = 17$, $n_{B+C} = 8$, $n_{B+R} = 8$, $n_{B+M} = 12$; Female–HFHS: $n_B = 20$, $n_{B+C} = 10$, $n_{B+R} = 7$, $n_{B+M} = 10$).

$n_{B+M} = 10$). **F** Representative WAT H&E-stained sections capture at 20X for each colonization group on SD or HFHS. **G** Proportion of adipocytes within size ranges 5–40 μm , 40–70 μm , >70 μm (SD: $n_B = 26$, $n_{B+C} = 11$, $n_{B+R} = 13$, $n_{B+M} = 13$; HFHS: $n_B = 25$, $n_{B+C} = 10$, $n_{B+R} = 7$, $n_{B+M} = 11$). Data represents mean $\pm$ SEM. Significant differences defined by Two-way ANOVA with Dunnett's multiple comparisons test relative to B only group within either diet. **C–E, G** Outliers removed with ROUT ($Q = 1\%$). **A–E** B = gray; B + C = orange; B + R = pink; B + M = purple. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. Source data are provided as a source data file; n indicates number of mice sampled. SD standard diet, HFHS high-fat-high-sucrose diet, gWAT gonadal white adipose tissue.

Bacterial compositional changes were also detected in mice fed HFHS. Alpha-diversity remained unchanged between B only and B + C or B + R, though was increased in B + M mice on HFHS (Fig. S1C). In all groups, there was a significant effect of diet on alpha-diversity with an overall reduction in HFHS-fed mice (B + C: $p = 0.020$, B + R: $p = 0.007$, B + M: $p < 0.001$; Fig. S1C). *C. albicans* influenced the greatest number of bacterial strains with an increase in *B. animalis* YL2 and *Enterocloster clostridioforme* YL32 and a decrease in *A. muris* KB18, *B. pseudococcoides* YL58, *E. faecalis* KB1, *F. plautii* YL31 and *T. muris* YL45 (Figs. 2I and S1D). *R. mucilaginosa* colonized mice displayed increased *B. animalis* YL2 and decreased *E. faecalis* KB1, while *M. restricta* was associated with increased *C. innocuum* I46 and *E. faecalis* KB1, and

decreased *Akkermansia muciniphila* YL44 (Figs. 2I and S1D). Altogether, exposure to a single fungal species early in life had long-term consequences for the bacterial community composition and its response to a diet high in fat and sucrose. These cross-kingdom alterations may provide an avenue for fungi to influence host metabolism.

Early-life fungal colonization has long-lasting consequences for the metabolic output of the gut microbiome in a diet-specific manner

Given the alterations to the gut microbiome observed on both diets with early-life fungal exposure, we explored whether there were also

functional alterations with fecal untargeted metabolomics. Similar to the alterations in bacterial community structure, the overall fecal metabolome was strongly influenced by diet (Fig. 3A–C and Table S3). Significant metabolome shifts were also found with early-life fungal exposure in B + C and B + R, but not B + M mice (Fig. 3A–C and Table S3). In line with this, the concentrations of specific metabolites were significantly shifted in B + C and B + R relative to B-only mice. *C. albicans* colonization resulted in the greatest number of significantly altered metabolites, with 8 increased and 6 decreased metabolites on SD, and 6 increased and 1 decreased metabolite on HFHS (Fig. 3D–G and Table S4). Several of these metabolites have been previously associated with *C. albicans*, including being produced (cystathionine, hypotaurine, arabinol), consumed (glutamine), degraded (urate) or simply associated with its presence^{37–42}, highlighting its capacity to contribute to the fecal metabolome. Some metabolites also displayed significant fold change with *R. mucilaginosa* colonization on SD only, where there were 5 increased and 4 decreased metabolites, including mannitol and glucose, which are known to be consumed by this yeast⁴³ (Fig. 3H, I and Table S4). There were no significantly altered metabolites in B + M mice on either diet. As this was a DIO model, we were also interested in the total lipid content remaining in the feces. There were no differences in fecal lipid concentration between B only and B + C or B + M mice on either diet (Fig. S2A, B). We did, however, find an increase in fecal lipid concentration in B + R mice on both diets (Fig. S2C). These data indicate that fungal colonization had a persistent and substantial impact on the fecal metabolome, including luminal nutrient content in mice.

To gain a better understanding of the functional consequences of fungal-induced alterations to the bacterial community, we constructed a community metabolic model by integrating the datasets from 16S rRNA amplicon sequencing and metabolomics. We used Model-based Integration of MIMOSA2 (Metabolite Observations and Species Abundances Version 2), which infers putative metabolites governed by the microbiome and the species that contribute to the experimental variation in metabolite levels⁴⁴. Given our gnotobiotic approach, we created a custom AGORA (Access to Global Online Research in Agriculture) database from the whole genome sequences of the OligoMM12 strains to assess the metabolic output of the exact strains present in the consortium.

Through this analysis, we found that each fungal species induced alterations in bacteria-governed metabolites on either diet. In B + C mice, uracil concentrations on SD and succinate on HFHS were significantly correlated with community metabolic potential, with eight of the OligoMM12 contributing to the variance, including strains whose relative abundances were shifted in B + C mice (*A. muris* KB18, *L. reuteri* I49 and *B. pseudococcoides* YL58 on SD, *T. muris* YL45 on HFHS; Figs. 2H, I and S2D). This suggests that differential levels of uracil and succinate occurred via *C. albicans*-induced shifts to the bacterial microbiome. In B + M mice, succinate concentration on SD and uracil concentration in HFHS were significantly correlated with community metabolic potential, with seven OligoMM12 taxa contributing to the variance in the levels of these metabolites (Fig. S2E). In this case, exposure to *M. restricta* was related to shifts in the relative abundance of *C. innocuum* I46 and *E. faecalis* KB1 (Fig. 2H, I). Lastly, in B + R mice, xanthine and hypoxanthine concentrations on SD and fumarate and taurine concentrations on HFHS were significantly correlated with community metabolic potential (Fig. S2F). Coinciding with their contributions to the variance of these metabolites, *R. mucilaginosa* colonization shifted the relative abundance of *L. reuteri* I49 on SD and *E. faecalis* KB1 on both diets (Fig. 2H, I), also suggesting this fungal species indirectly influenced the microbiome's metabolic output via changes in bacterial species abundance and/or metabolism. In combination with our ecological analysis via 16S rRNA amplicon sequencing, community-metabolic modeling highlights interkingdom interactions as driving forces of the composition and functional output of the gut microbiome.

Early-life fungal colonizers modulate diet-induced obesity and metabolic disease development

We next evaluated whether early-life fungal exposure was causally involved in host energy metabolism and the metabolic response to the HFHS diet. For this, we carried out an extensive assessment of metabolic health, which included measurements of body and tissue weight, body composition, adipocyte size, glycemic response to oral glucose, plasma lipids and metabolic hormones.

Given the known effect of sex on metabolic measurements and DIO⁴⁵, we designed this study to identify sex-driven effects on metabolism. A sex effect for body weight was identified with repeated measures (RM)-ANOVA; therefore, males and females were analyzed separately (sex $p < 0.001$). In males, both colonization and time had a significant effect, though the effect of diet was not significant (colonization $p < 0.001$, diet $p = 0.141$, time $p < 0.001$). Similar outcomes were found in females (colonization $p < 0.001$, diet $p = 0.397$, time $p < 0.001$). Area under the curve (AUC) for body weight from 3–12 weeks was significantly lower in B + C mice compared to the B only group on both diets in males and on HFHS in females (Fig. 4A, B). Male B + C mice had lower body weight at every week compared to the B only group regardless of diet, while females had lower body weight at weeks 4, 5 and 9 on SD diet, and lower weight at all time points on HFHS (Fig. 4A, B). In contrast, we detected higher body weight in female SD-fed mice exposed to *M. restricta* at weeks 10–12 (Fig. 4B). *R. mucilaginosa* did not drive changes in body weight in either sex. We identified reciprocal findings in body composition, measured by time-domain nuclear magnetic resonance (TD-NMR). On SD, *C. albicans* resulted in lower fat mass, while *R. mucilaginosa* and *M. restricta* resulted in higher adiposity compared to the B only group (Fig. 4C). Higher adiposity in B + M mice on SD was consistent with greater weight of the perigonadal fat pad (primary visceral fat pad in rodents⁴⁶) in both sexes compared to B only, and a higher proportion of large adipocytes (>70 μm diameter) in this depot (Fig. 4D–G). Expansion of visceral adiposity, along with adipocyte hypertrophy are determinants of metabolic health²⁸, highlighting the ability of core fungal species to differentially regulate host energy storage in a consequential manner for metabolic disease development.

In obesogenic diet conditions (HFHS), mice colonized with *C. albicans* failed to gain body fat mass and peri-gonadal fat pad weight (Fig. 4C–E). Instead, these mice displayed a higher lean mass percentage compared to mice harboring bacteria alone (Fig. S3A, B). This resistance to DIO in mice colonized with *C. albicans* was also accompanied by histological changes in WAT. Compared to B only mice, B + C mice displayed a lower proportion of large adipocytes (>70 μm diameter) and a greater percentage of small or medium size adipocytes on either diet, suggesting reduced lipid storage in this depot, which expands in response to DIO⁴⁷ (Fig. 4F, G). In addition, *C. albicans* colonization resulted in lower liver, kidney and heart weight on HFHS in both sexes (Fig. S3C–H), as well as lower kidney and heart weight in males on SD conditions (Fig. S3E, G). Increased weight of these organs has been characterized as markers of metabolic disease development^{48–50}, further supporting that *C. albicans* promoted systemic resistance to DIO. Importantly, we did not detect differences in energy intake driven by fungal colonization on either diet, except for an increase in B + C mice at 9 weeks on HFHS (Fig. S3I, J). This indicates that the stark changes in body and tissue weight, and adiposity, were not a result of changes in energy intake and occurred in both standard and obesogenic diet conditions.

Given the overt signs of altered metabolic states driven by early-life fungal exposures, we further explored metabolic outcomes. To evaluate the effect of fungal colonization on glucose metabolism, we performed an oral-glucose tolerance test (OGTT) at 12 weeks of age. A sex effect was identified with RM-ANOVA, therefore males and females were analyzed separately (sex $p < 0.001$). The effect of colonization,

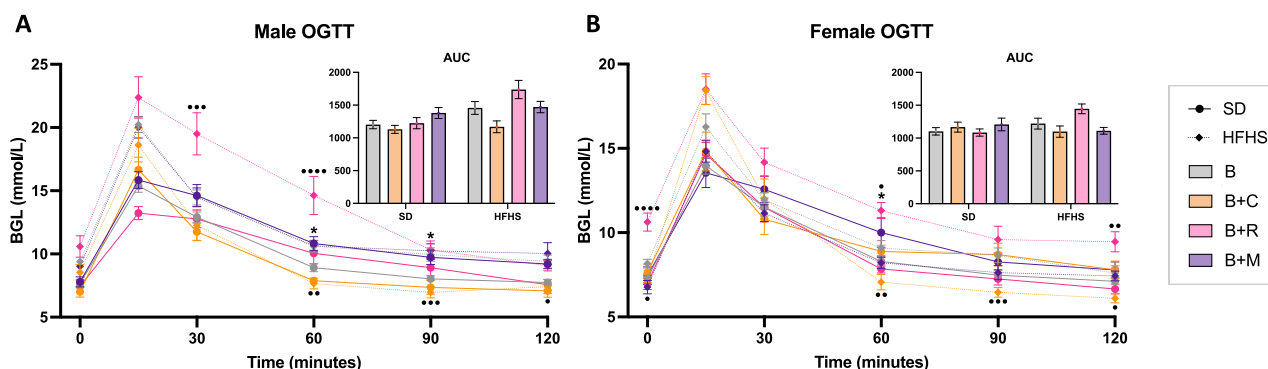


Fig. 5 | Gut fungi modulate oral glucose tolerance. A, B Oral glucose tolerance test (OGTT) with trapezoidal total glucose area under the curve (AUC) in males and females, respectively (Male–SD: $n_B = 23$, $n_{B+C} = 8$, $n_{B+R} = 10$, $n_{B+M} = 12$; Male–HFHS: $n_B = 22$, $n_{B+C} = 9$, $n_{B+R} = 8$, $n_{B+M} = 12$; Female–SD: $n_B = 17$, $n_{B+C} = 8$, $n_{B+R} = 8$, $n_{B+M} = 12$; Female–HFHS: $n_B = 20$, $n_{B+C} = 12$, $n_{B+R} = 7$, $n_{B+M} = 10$). Data represents mean $\pm$ SEM. Significant differences defined by Two-way ANOVA with Dunnett's multiple

comparisons test relative to B only group within either diet. B = gray; B + C = orange; B + R = pink; B + M = purple. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. Significant differences indicated by asterisk for SD and dots for HFHS. Source data are provided as a source data file; n indicates number of mice sampled. SD standard diet, HFHS high-fat-high-sucrose diet.

diet and time were significant in both males (colonization $p < 0.001$, diet $p < 0.001$, time $p < 0.001$) and females (colonization $p = 0.014$, diet $p = 0.002$, time $p < 0.001$). In both sexes and in HFHS diet conditions, colonization with *R. mucilaginosa* resulted in elevated blood glucose levels (BGL) at several time points, including 30 and 60 min in males, as well as baseline, 60 and 120 min in females (Fig. 5A, B). Meanwhile, B + M mice fed SD displayed higher BGL at 60 min in both sexes along with 90 min in males (Fig. 5A, B). In contrast, B + C mice consuming HFHS displayed lower BGL at 60 and 90 min in both sexes, along with 120 min in females (Fig. 5A, B). While the effect of fungal colonization was not reflected in total glucose area under the curve (AUC) measurements (Fig. 5A, B), alterations at individual time points suggest that intestinal fungi can influence host glucose control.

Measurement of metabolic hormones and metabolically relevant cytokines revealed endocrine alterations consistent with the above findings. Similar to changes in adiposity, levels of the satiety hormone, leptin, were elevated in B + M mice on both diets compared to B only, and lower in B + C mice fed HFHS, independently of sex (Fig. S4A). The hunger-stimulating hormone, ghrelin, was also higher in B + M mice on both diets in males and in HFHS fed females (Fig. S4B, C), along with interleukin (IL)-6 in male mice fed SD and glucagon-like peptide 1 (GLP-1) in HFHS fed mice, independently of sex (Fig. S4D–F). GLP-1 was also higher in B + R mice fed HFHS compared to B only (Fig. S4F). Fungal colonization did not influence the concentrations of monocyte chemoattractant protein-1 (MCP-1), glucagon or peptide tyrosine tyrosine (PYY) (Fig. S4G–K). These results suggest that fungal-induced metabolic changes occur at least in part through modulation of the host endocrine system.

Fungal colonization also impacted systemic lipid profiles. Male B + C mice displayed lower plasma triglycerides on both diets and both sexes had lower total cholesterol when fed HFHS, reflective of their lean phenotype and resistance to DIO (Fig. 6A–D). In contrast, B + R mice displayed higher triglycerides in females, low-density lipoprotein (LDL) cholesterol in males fed HFHS and in females on both diets, alongside lower HDL cholesterol in females fed HFHS (Figs. 6A–F and S4L, M). We also found that B + R mice on HFHS displayed enhanced hepatic lipid accumulation in both sexes compared to B only (Fig. 6G–I). These alterations in systemic lipid profiles and hepatic lipid accumulation induced by *R. mucilaginosa* colonization are indicative of dyslipidemia and metabolic disease. Interestingly, both B + C and B + M female mice displayed lower hepatic lipid droplets, despite showing different metabolic phenotypes (Fig. 6G–I). Altogether, we show that each of these core human mycobiome species distinctly modulate several aspects of host

metabolism and metabolic disease development under normal and obesogenic dietary conditions.

Fungal colonization does not influence nutrient transporter expression

To better understand the mechanisms involved in these phenotypes, we sought to determine whether alterations to the fecal metabolome driven by fungal colonization may influence intestinal nutrient absorption. To do this we employed murine jejunal enteroid-derived monolayers treated with fecal filtrates from mice colonized with bacteria alone or in combination with each fungal species. We then quantified the expression of *Cd36* and *Slc5a1*, to examine sugar and lipid absorption capacity, respectively. Given the induction of mucosal type 17 responses by gut fungi⁵¹, we also treated the monolayers with IL-17 as previous work has shown this cytokine to be involved in the expression of *Cd36*⁵². There were no changes to the expression of *Cd36* and *Slc5a1* based on colonization status of the mice from which the fecal filtrates were derived, however IL-17 had a significant effect on both genes (*Cd36*: $p = 0.0119$, *Slc5a1*: $p = 0.0149$; Fig. S5). These findings suggest that metabolome alterations driven by each fungal species did not significantly contribute to changes in small intestinal nutrient absorption.

Fungal colonizers modulate the visceral adipose tissue immune landscape

Immune cell populations in adipose tissue are critical regulators of tissue function with implications for metabolic health⁵³. Therefore, we evaluated the impact of early-life exposure to these three fungal species on the visceral WAT immune landscape. Among the different fat depots, visceral WAT has the greatest inflammatory potential and has been studied extensively in the context of microbiome-mediated alterations to adipose tissue immunity^{28,54}. We carried out a comprehensive assessment of 27 myeloid and lymphoid immune cell populations in peri-gonadal fat tissue and in the spleen at 12 weeks of age (Fig. S6 and Table S5). Comparing the immune landscape of WAT to that of the spleen allowed us to discern immune changes specific to this depot from systemic ones. This analysis yielded numerous significant changes in WAT immune cell populations that varied according to fungal colonization, diet and sex.

Similar to the potent ability of *C. albicans* to modulate host immunity at distal sites, such as the lungs²⁶, *C. albicans* colonization resulted in drastic changes to the visceral WAT innate and adaptive immune landscape, compared to mice harboring bacteria only. Under SD diet conditions, B + C mice displayed higher numbers of

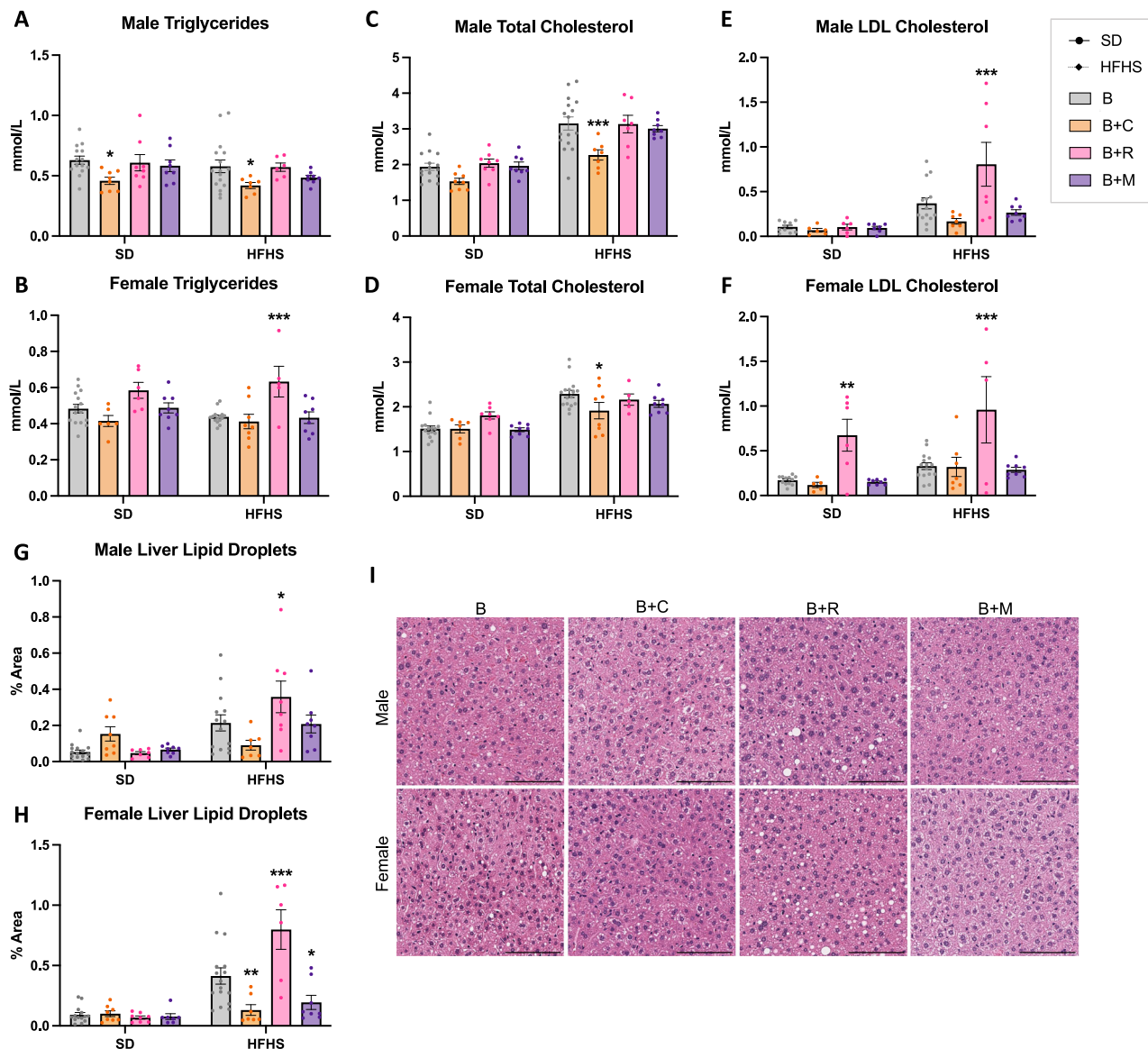


Fig. 6 | Gut fungi differentially modulate dyslipidemia and hepatic lipid accumulation. **A, B** Plasma triglycerides in males and females, respectively (Male—SD: $n_B = 15$, $n_{B+C} = 8$, $n_{B+R} = 8$, $n_{B+M} = 8$; Male—HFHS: $n_B = 16$, $n_{B+C} = 7$, $n_{B+R} = 7$, $n_{B+M} = 8$; Female—SD: $n_B = 15$, $n_{B+C} = 6$, $n_{B+R} = 7$, $n_{B+M} = 8$; Female—HFHS: $n_B = 16$, $n_{B+C} = 5$, $n_{B+R} = 8$, $n_{B+M} = 8$). **C, D** Plasma total cholesterol in males and females, respectively (Male—SD: $n_B = 16$, $n_{B+C} = 8$, $n_{B+R} = 8$, $n_{B+M} = 8$; Male—HFHS: $n_B = 16$, $n_{B+C} = 7$, $n_{B+R} = 7$, $n_{B+M} = 8$; Female—SD: $n_B = 15$, $n_{B+C} = 6$, $n_{B+R} = 7$, $n_{B+M} = 8$; Female—HFHS: $n_B = 16$, $n_{B+C} = 8$, $n_{B+R} = 5$, $n_{B+M} = 8$). **E, F** Plasma LDL cholesterol in males and females, respectively (Male—SD: $n_B = 10$, $n_{B+C} = 6$, $n_{B+R} = 7$, $n_{B+M} = 7$; Male—HFHS: $n_B = 16$, $n_{B+C} = 7$, $n_{B+R} = 8$, $n_{B+M} = 8$; Female—SD: $n_B = 15$, $n_{B+C} = 6$, $n_{B+R} = 6$, $n_{B+M} = 8$; Female—HFHS: $n_B = 16$, $n_{B+C} = 7$, $n_{B+R} = 5$, $n_{B+M} = 8$). **G, H** Liver lipid droplets

quantified as percent area of histological images in males and females, respectively (Male—SD: $n_B = 16$, $n_{B+C} = 8$, $n_{B+R} = 8$, $n_{B+M} = 8$; Male—HFHS: $n_B = 15$, $n_{B+C} = 7$, $n_{B+R} = 8$, $n_{B+M} = 8$; Female—SD: $n_B = 15$, $n_{B+C} = 8$, $n_{B+R} = 8$, $n_{B+M} = 8$; Female—HFHS: $n_B = 17$, $n_{B+C} = 7$, $n_{B+R} = 6$, $n_{B+M} = 8$). **I** Representative liver H&E-stained sections captured at 20X for each colonization group on HFHS in males or females. Data represents mean $\pm$ SEM. Outliers removed with ROUT ($Q = 1\%$). Significant differences defined by Two-way ANOVA with Dunnett's multiple comparisons test relative to B only group within either diet. B = gray; B + C = orange; B + R = pink; B + M = purple. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. Source data are provided as a source data file; n indicates number of mice sampled. SD standard diet, HFHS high-fat-high-sucrose diet, LDL low-density lipoprotein.

$\gamma\delta$ T cells, innate lymphoid cell (ILC) 1, ILC3, intermediate macrophages (CD206^{lo/int}Ly6C⁺MHCII⁺), conventional dendritic cell (cDC) 1, CX3CR1⁺ DCs and neutrophils in both sexes compared to B only (Fig. 7A and Tables S6, 7). In males only, B + C mice displayed higher Th1, Th17, CD8⁺ T cells, B cells, NK cells, total CD64⁺ cells including CX3CR1⁺ macrophages, vascular-associated macrophages (VAM) type 1 and 2 (CD206^{hi}MHCII⁺Tim4^{int} and Tim4^{hi}, respectively) and preVAM (CD206^{lo/int}Ly6C⁺MHCII⁺CD11c⁻) (Fig. 7A and Tables S6, 7). Except for changes in Th1, Th17 and ILC3 in both sexes, and $\gamma\delta$ T cells, cDC1 and neutrophils in females, most of the shifts in WAT immune cell populations were not identified in the spleens of mice colonized

with *C. albicans* (Fig. S7A and Tables S8, 9), indicating that the broad and drastic effect on WAT immunity is not solely a result of systemic immune changes. These changes are also not a result of *C. albicans* infection or translocation from the gut since we could not detect the growth of this yeast in internal organs or blood.

Both *R. mucilaginosa* and *M. restricta* similarly influenced the populations of immune cells in WAT on a SD, although their effects were not as pronounced as those observed with *C. albicans*. In females, B + R mice displayed higher ILC2, CD11c⁺ macrophages, preVAM, VAM2 and eosinophils, with a parallel enrichment in the spleen for ILC2 only (Figs. 7A and S7A and Table S6–9). In B + M mice, there were fewer ILC2

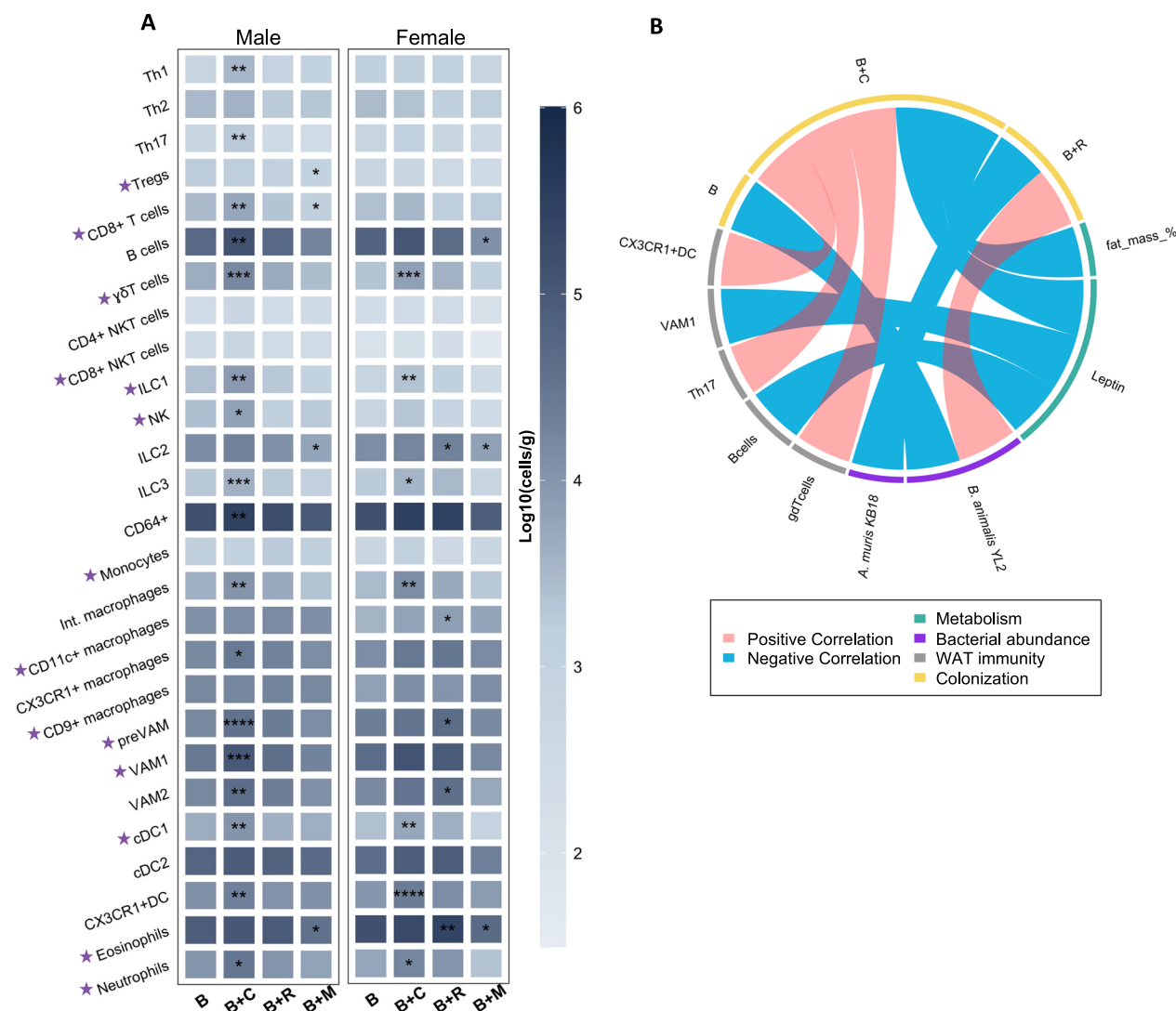


Fig. 7 | Fungi modulate the visceral WAT immune cell landscape, microbiome and metabolism in an integrated manner. A WAT immune cell populations in SD diet fed mice. Data represented as mean log10(cells/gram of tissue) (Male: $n_B = 23$, $n_{B+C} = 8$, $n_{B+R} = 10$, $n_{B+M} = 12$; Female: $n_B = 17$, $n_{B+C} = 8$, $n_{B+R} = 8$, $n_{B+M} = 8$). Outliers identified with ROUT ($Q = 1\%$). Sex effect assessed by Three-way ANOVA and $p < 0.05$ indicated with a star. Significant differences between fungal colonized and B only groups defined by Two-way ANOVA with Dunnett's multiple comparisons

test $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$. **B** Spearman correlation between variables grouped by metabolism, bacterial abundance, white adipose tissue (WAT) immunity and colonization ($n_B = 18$, $n_{B+C} = 14$, $n_{B+R} = 14$, $n_{B+M} = 8$). Correlations with coefficients $> |0.6|$ shown. Positive correlation = pink; negative correlation = blue, metabolism variables = green; bacterial abundance variables = purple; WAT immunity variables = gray; colonization variables = yellow. Source data are provided as a source data file; n indicates number of mice sampled.

and eosinophils in both sexes, fewer Tregs and CD8⁺ T cells in males only, and fewer B cells in females only compared to B only (Fig. 7A and Tables S6, 7). Importantly, none of these changes in population dynamics were mirrored in the spleen (Fig. S7A and Tables S8, 9), indicating that fungal colonizers can uniquely modulate the immune landscape within WAT, particularly under SD conditions, which carries significant implications for metabolic outcomes.

We next examined how fungi influenced this same immune landscape under obesogenic conditions. As observed in normal diet conditions, *C. albicans* elicited the most potent immune modulation in WAT. B + C mice displayed more cDC1 and neutrophils in both sexes, Th17, CD8⁺ T cells, B cells, $\gamma\delta$ T cells, ILC1, ILC2, VAM1 and eosinophils in males only, and lower CD9⁺ macrophages in females only (Fig. S8A and Tables S10, 11). We also observed alterations to Th17 and ILC1 in the spleen (Fig. S7B and Tables S12, 13), suggesting that mice fed HFHS experience systemic immune changes that may also impact WAT. Fewer populations were modulated in B + R mice, including higher ILC2, ILC3, VAM1 and VAM2 in males only, with no parallel shifts in the

spleen (Figs. S7B, S8A and Tables S10–13). Lastly, B + M mice displayed lower Th17 in both sexes (also seen in the spleen for males), and lower Th1 and ILC3 in females (Figs. S7B, S8A and Tables S10–13). Together, this extensive evaluation of WAT immunity indicates that early-life fungal colonization can uniquely and causally modulate the immune landscape of this important fat depot in both standard and obesogenic diet conditions, with differential effects driven by sex.

Systems-level integration show fungal associations with microbiome, metabolic and immune changes

To gain insights into the complex connections between the biological processes measured in our study, we carried out an integrative analysis based on Spearman correlations between microbial, metabolic and immune datasets. This type of unsupervised analysis can highlight interrelationships between the involved biomolecules and their functions⁵⁵. Notably, we found that colonization with *C. albicans* displayed most of the detected significant correlations in mice fed SD (Fig. 7B). This analysis revealed significant negative correlations

between *C. albicans* colonization and fat mass and leptin, reflective of the lean phenotype. We also observed significant integration between *C. albicans* colonization and $\gamma\delta$ T cells, Th17 and CX3CR1+ DCs (Fig. 7B), in line with the increased type 17 inflammation in the WAT of these mice. Meanwhile, *R. mucilaginosa* colonization was positively correlated with the relative abundance of *B. animalis* YL2, contrasting with mice colonized with only bacteria, and negatively correlated with *A. muris* KB18 (Fig. 7B). On HFHS, *C. albicans* colonization maintained a negative correlation with fat mass and leptin, along with body weight at 12 weeks (Fig. S8B). Overall, these findings suggest that *C. albicans* influences both metabolic programming and the inflammatory tone of WAT, which may regulate host energy storage. On the other hand, *R. mucilaginosa* may elicit its effects on host metabolism through its relationship with the ecology of the bacterial microbiome.

Discussion

Using a gnotobiotic DIO model, we determined that three fungal species previously linked to childhood BMI impact host energy balance and response to an obesogenic diet in a causal manner. Our approach identified striking immunometabolic effects of *C. albicans* colonization, resulting in a lean phenotype and resistance to obesity development, alongside broad WAT inflammation. In contrast, *R. mucilaginosa* and *M. restricta* drove increased adiposity under distinct metabolic phenotypes and WAT immune landscapes, with *R. mucilaginosa* colonization causing hyperglycemia, dyslipidemia and shifts in WAT immunity that reflect enhanced obesity development. In all cases, fungal colonization induced broad and long-lasting shifts in microbiome composition and function, revealing interkingdom ecological interactions as potential mediators of host energy metabolism. As the multikingdom nature of the gut microbiome is increasingly recognized, our work provides evidence of fungal species as causal contributors to the interaction between the gut microbiome and host metabolism.

The distinct metabolic phenotypes observed in mice fed control and obesogenic diets may be related to the metabolic abilities exhibited by each yeast species. Specifically, in the context of excess adiposity, *R. mucilaginosa* and *M. restricta* demonstrate unique metabolic functions related to lipids. *R. mucilaginosa* is considered an oleaginous yeast with lipids constituting up to 50% of the cell's dry weight⁵⁶. Additionally, certain *Rhodotorula* strains can convert inulin into polyol fatty acids in substantial quantities⁵⁷. While the HFHS diet did not include inulin, the possibility of other fibers included in the diet (i.e., cellulose) being converted to fatty acids could account for the surplus of lipids detected in stool (Fig. S2A–C). Although it has not been established whether this yeast is able to enhance lipid harvest in the gut lumen, it is plausible that its synthesis of long-chain fatty acids could increase host energy intake and drive excess lipid accumulation in WAT, resulting in signs of metabolic syndrome, especially dyslipidemia^{56,58}. WAT immunity in these mice displayed obesity-associated shifts consistent with reports in other DIO models, such as increased CD11c⁺ macrophages and VAM2 (Figs. 7A and S8A)^{59,60}. Interestingly, the immune landscape in *R. mucilaginosa* colonized mice also included cell populations linked to a more favorable immune milieu and a lean phenotype, such as ILC2 and eosinophils (Figs. 7A and S8A)⁶¹. This may reflect a compensatory shift in these populations in an attempt to promote WAT adaptation to the early stages of adipose tissue expansion during the onset of obesity.

Meanwhile, *Malassezia* is a lipid-dependent genus typically associated with sebaceous areas of the skin and is a core member of the breast milk mycobiome, providing a route of transmission to breastfed infants^{62,63}. The inability of *M. restricta* to stably colonize gnotobiotic mice excludes the possibility that this yeast could directly influence lipid harvest from the diet. Instead, it is more likely that its impact on the bacterial microbiome influenced early-life immune and metabolic programming, resulting in increased adiposity later in life. Future

studies should examine whether this is dependent on exposure during the early-life window by inoculating mice at different time points. Intriguingly, B+M mice failed to develop metabolic disease in the presence of increased adiposity (Figs. 4–6), suggesting an enhanced ability to expand their fat depots in a controlled manner. It is possible that this appropriate WAT expansion is in connection with immune regulation of adipose tissue, given the reduced inflammatory populations in WAT relative to mice harboring bacteria only (Figs. 7A and S8A). While intestinal *M. restricta* has been associated with inflammatory bowel disease due to its influence on the mucosal immune system, the potential for similar immune modulation at distal sites, such as WAT remains plausible^{64,65}.

The most striking finding from our work is the lean and obesity-resistant phenotype of *C. albicans* colonized mice (Figs. 4–6). Our findings support previous work showing a reduction in body weight in adult mice fed a high-fat diet mixed with live *C. albicans* cultures¹⁸, and suggest that the obesity-resistant effect is stronger if colonization with this yeast occurs at birth. The mechanisms behind this phenotype are yet to be elucidated but likely involve WAT immune responses. While these mice did not have a disseminated infection, this enhanced inflammatory response and immune system activation may pose an increased energetic burden on the host⁶⁶. *C. albicans* is a pathobiont that strongly modulates mucosal, systemic, and distal immune responses, which are critical for the host-microbe relationship involved in keeping this fungus contained to the gastrointestinal tract²⁶. However, because this inflammatory response persisted even under a lower fungal burden on HFHS diet, it is likely that other mechanisms aside from the energetic demand for host defense are at play. Additional mechanisms involved in energy storage should be explored in future studies, including WAT remodeling and thermogenesis⁶¹.

The strong shifts in stool metabolites in *C. albicans* colonized mice (Fig. 3) may reflect direct and indirect contributions of yeast metabolism in the gut. Interestingly, some of the metabolites found elevated in these mice have been associated with leanness. D-arabitol increased bacterial production of short-chain fatty acids to enhance WAT browning in a model of DIO⁶⁷. Oral glutamine supplementation reduced waist circumference and serum insulin in individuals classified as overweight and reduced adiposity and improved insulin sensitivity in high-fat diet-fed rats⁶⁸, and succinate treatment during DIO has been linked to increased mitochondrial biogenesis in skeletal muscle⁶⁹. Meanwhile, urate, which was lower in *C. albicans* colonized mice (Fig. 3), has been positively correlated with BMI and fat mass in youth^{70,71}. Interestingly, the fecal metabolome did not impact nutrient transporter expression in jejunal enteroid-derived monolayers (Fig. S5), suggesting that these metabolite alterations may be involved in mechanisms beyond energy intake such as immune regulation at distal sites⁷². However, it is unclear whether these metabolites are directly produced by *C. albicans* or the consequence of interkingdom interactions. Studies examining the effect of the gut metabolome in monocolonized mice may provide further clarity. Overall, these data suggest that there are direct and indirect metabolic consequences of *C. albicans* colonization that may be involved in mediating the lean and resistant to obesity phenotype displayed by these mice.

Our gnotobiotic model enabled us to determine the causal impact of early-life fungal exposures, though there are limitations to consider. While our model intended to capture obesity development initiated through Westernized diet consumption during youth and into adulthood, this was a relatively short time frame for a DIO model. We did not achieve the full manifestation of metabolic syndrome, which is typically seen after 8–12 weeks on this type of diet if initiated during adulthood⁷³, but it is unclear if this is a feature of our gnotobiotic approach, the reduced timeline or both. Nevertheless, various markers of metabolic disease emerged allowing us to evaluate metabolic phenotypes.

The Oligo-MM12 consortium, while substantially resembling a typical mouse gut microbiome, only encompasses 66.6% of the metabolic functional units (KEGG modules) found in conventional mice³³. Thus, our simplified model lacks representation of several aspects of microbial metabolism and interkingdom dynamics. Further, like all mouse models, our model does not replicate the many ecological relationships occurring in the human microbiome⁷⁴. We previously reported correlations between *C. albicans*, *R. mucilaginosa* and *M. restricta* with infant BMI scores²², though the directionality of these relationships contrasts with the phenotypes observed here in mice. Additionally, it was recently reported that Oligo-MM12 mice display several differences in energy metabolism from conventional mice, including fat mass, respiratory exchange ratio and liver metabolome⁷⁵. Overall, there are likely complex inter- and intra-kingdom interactions involved in fungal regulation of host metabolism that should be further explored in more complex microbial communities and other animal models, such as rats, which display greater microbiome similarity to that of humans⁷⁶. Additionally, while our jejunal enteroid model allowed us to examine the direct influence of gut metabolites under fungal colonization conditions on nutrient transporter expression, this model lacked the complexities of the gut microenvironment. Although we included IL-17 to provide relevant immune signaling, it is likely that other cytokines and cell types are involved. Future studies should aim to incorporate the various components that may be involved in mediating changes in nutrient uptake, such as immune cells, vasculature, neuro-endocrine signaling and direct microbial interactions with host cells⁷⁷. Lastly, it is important to consider the impact of the maternal microbiome on offspring development. In our model, female breeders were treated with the fungal species at the onset of the breeding process. The maternal gut microbiome influences the offspring's gut microbiome, immune and metabolic development^{78,79}; therefore, it is possible that the fungal treatment during gestation may also play a role in the observed phenotypes. Future work should explore whether these outcomes were dependent on gestational factors.

In summary, our research carries significant implications for advancing our understanding of the complex interplay between the gut microbiome and host metabolism. It sheds light on the previously underexplored role of fungal species in shaping host energy balance and responses to obesogenic diets. These findings not only emphasize the importance of considering the multi-kingdom nature of the gut microbiome but also suggest that fungal species play a significant role in modulating host metabolism. This work provides a foundation for deeper exploration of fungi-host interactions, the underlying mechanisms, and their potential implications on metabolic diseases, offering new avenues for research in this emerging field.

Methods

Gnotobiotic mouse model

Seven to thirteen-week-old, germ-free C57Bl/6J mice were obtained from the International Microbiome Center (IMC) gnotobiotic mouse facility at the University of Calgary. Female mice were orally gavaged twice, 2 days apart, with 100 µl of a consortium of organisms, that was prepared in house. These consortia included (1) the Oligo-MM12³³ alone “B” (*A. muris* KB18, *A. muciniphila* YL44, *B. caecimuris* I48, *B. animalis* YL2, *B. pseudococcoides* YL58, *C. innocuum* I46, *E. clostridioforme* YL32, *E. faecalis* KB1, *F. plautii* YL31, *L. reuteri* I49, *M. intestinale* YL27, and *T. muris* YL45) or (2) in combination with *C. albicans* (This strain was isolated from infant stool samples enrolled in a clinical study at the Alberta Children's Hospital in Calgary, Canada⁸⁰) “B + C”, (3) *R. mucilaginosa* (DSMZ; DSM 70825) “B + R”, or (4) *M. restricta* (ATCC; MYA4611) “B + M”. Experiments were completed with groups B, B + C and B + R run simultaneously, or B and B + M run simultaneously. Each colonization group was kept in a separate gnotobiotic isolator. Inoculum were prepared under anaerobic conditions by combining

100 µl of 3-day-old cultures of each bacterial species, 1-day-old culture of *C. albicans*, 2-day-old culture of *R. mucilaginosa* and 7-day-old culture of *M. restricta*. Bacteria were grown in fastidious anaerobe broth at 37 °C (Lab M Limited, UK); *C. albicans* and *R. mucilaginosa* were grown in yeast-mold broth at room temperature (YM; Becton Dickinson, USA); and *M. restricta* was grown in modified Dixon media (mDixon)⁸¹ at 30 °C. mDixon was prepared in house by combining 36 g/L malt extract (Becton Dickson), 20 g/L desiccated Ox-bile (Sigma-Aldrich, USA), 10 mL/L tween-40 (Sigma-Aldrich), 6 g/L peptone (Sigma-Aldrich), 2 mL/L glycerol (VWR), 2 mL/L Oleic Acid (Sigma-Aldrich), and 15 g/L agar (Thermo Fisher, USA), and was adjusted to pH 6. After the initial gavage, females were paired with males for mating at a 2:1 ratio, respectively. To support early-life exposure to fungal species, fungal cultures were spread on the dams' abdomen and nipple regions twice on days 2–6 after birth⁸². Mice were kept at a maximum of five animals per cage, under a 12-h light/dark cycle, 40% relative humidity, 22–25 °C and ad libitum access to sterile food and water. All experiments were conducted under protocols approved by the University of Calgary Animal Care Committee, following the guidelines of the Canadian Council on Animal Care (AC20-0176).

DIO model

At 21-days of age, mice were weaned onto a powdered SD (LabDiet JL Rat and Mouse/Auto 6F 5K52, Richmond, IN, USA) or HFHS diet (HFHS; DYET#1013915, Bethlehem, PA, USA), and remained on the diets until the experimental endpoint at 12 weeks of age (Males–SD: $n_B = 23$, $n_{B+C} = 8$, $n_{B+R} = 10$, $n_{B+M} = 12$; Females–SD: $n_B = 18$, $n_{B+C} = 8$, $n_{B+R} = 8$, $n_{B+M} = 12$; Males–HFHS: $n_B = 22$, $n_{B+C} = 10$, $n_{B+R} = 8$, $n_{B+M} = 12$; Females–HFHS: $n_B = 20$, $n_{B+C} = 12$, $n_{B+R} = 7$, $n_{B+M} = 10$). To ensure sterility, the HFHS diet was irradiated (5–20 kGy) and both diets were autoclaved at 132 °C for 30 min. Mice had ad libitum access to food and water, and food intake was measured at 3, 5, 7 and 9 weeks of age as the average difference in food jar weight over 3 consecutive days. Body weight was measured weekly and whole-body fat and lean mass were measured at 12 weeks of age, following export from gnotobiotic isolators by time-domain nuclear magnetic resonance (TD-NMR) with the Minispec LF90II (Bruker, USA). Wet weight of peri-gonadal adipose tissue, liver, kidney and heart was determined at the experimental endpoint.

Quantification of viable fungi

Fecal fungal load was first assessed by culturing samples collected at 6–7 weeks of age on selective medium. Mouse fecal pellets were weighed and homogenized in 500 µl of sterile phosphate-buffered saline (PBS; Corning, USA) at 15 Hz, for 2 min. Homogenates were serially diluted in sterile PBS. Dilutions were spread on YM (B + C, B + R) or mDixon (B + M) agar plates supplemented with gentamycin (10 µg/ml; Sigma-Aldrich) and chloramphenicol (200 µg/ml; Sigma-Aldrich). Plates were incubated at room temperature for 3 (YM) or 10 (mDixon) days before colonies were counted to determine viable fungal concentration. Blood along with kidney and liver homogenates (homogenized in 1 mL of sterile PBS with a steel bead at 30 Hz, for 3 min) from B + C mice were also plated as described above.

gDNA isolation from fecal samples

Genomic template DNA was isolated from mouse fecal samples at the specified time points with the DNeasy PowerSoil Pro kit (Qiagen, Germany) according to the manufacturer's instructions with an added cell lysis step at TissueLyser II bead beater (Qiagen) and eluted in nuclease-free water. Samples were stored at –20 °C until further processing for quantitative polymerase chain reaction (qPCR) or 16S rRNA amplicon sequencing.

Fungal qPCR

Fungal DNA concentration in fecal samples at 12 weeks of age was determined by quantitative amplification of the fungal ITS spacer

region⁸³. All qPCR reactions were carried out with the StepOne Plus Real-Time PCR System (Applied Biosystems, USA). Reactions were performed at 20 µl, consisting of 10 µl of iQ SYBR Green Supermix (BioRad Laboratories, USA), 100 nM ITS1F (5'-CTTGGTCATTAGAG GAAGTAA-3'), 500 nM ITS4 (5'-TCTCTCCGCTTATTGATATGC-3'), 1 ng of template DNA and topped up with nuclease-free water. The thermocycler program consisted of an initial 5-min step at 95 °C, followed by 32 cycles of 95 °C for 30 s, 55 °C for 30 s, 72 °C for 1 min, and a final melt curve of 95 °C for 15 s, 60 °C for 1 min, 95 °C for 15 s and 60 °C for 15 s. All samples were run in duplicate, and concentration was calculated using a standard curve of known concentration of *C. albicans* gDNA.

16S rRNA amplicon sequencing

DNA samples were sent to Microbiome Insights (Vancouver, Canada) where PCR was used to amplify the V4 region of the 16S rRNA gene with 515F/806R primers⁸⁴ using Phusion Hot Start II DNA Polymerase (Thermo Scientific). Samples from each experiment (experiment 1: B, B + C and B + R; experiment 2: B and B + M) were sequenced in the same run. PCR reactions were cleaned up, normalized using the high-throughput SequalPrep Normalization Plate Kit (Applied Biosystems), and quantified accurately with the KAPA qPCR Library Quantification kit (Roche, Switzerland). This generated ready-to-pool, dual-indexed amplicon libraries⁸⁵. Controls without template DNA and mock communities with known amounts of selected bacteria or fungi were included in the PCR and downstream sequencing steps to control for microbial contamination. The pooled and indexed libraries were denatured, diluted, and sequenced in paired-end modus on an Illumina MiSeq (Illumina Inc., USA).

Amplicon sequence processing

Sequence processing was conducted in R v.4.1.1⁸⁶. Sequences were checked for quality, trimmed, merged, and checked for chimeras using the DADA2 v1.20.0 pipeline⁸⁷. Taxonomy was assigned as amplicon sequence variants (ASVs) using a custom build database, containing the 16S rRNA amplicon sequences of the Oligo-MM12 strains⁸⁸. Any ASVs that did not align with the Oligo-MM12 strains were assigned as "Other". The ASV table was preprocessed using the *Phyloseq* package v.1.36.0⁸⁹. Overall, 191 unique ASVs were detected. Samples with less than 1000 sequencing reads were excluded and ASVs appearing in only one sample were removed. The remaining dataset was filtered for ASVs appearing at least three times in 20% of samples, leaving 15 unique ASVs. This dataset was used for all subsequent analyses.

Metabolomics processing

Fecal samples collected at the 12-week endpoint were processed for untargeted metabolomics. Samples from each experiment (experiment 1: B, B + C and B + R; experiment 2: B and B + M) were analyzed in the same run. Samples were homogenized in 5 volumes of 50% ice-cold methanol (mg:ml) at 30 Hz for 3 min, followed by a 30-min incubation on ice and spun at max speed for 10 min at 4 °C. Supernatant was removed and incubated at -80 °C for up to 1 week before diluting 10X in 50% methanol. Samples were frozen at -80 °C and delivered to the Calgary Metabolomics Research Facility. General metabolomics runs were performed on a Q Exactive HF Hybrid Quadrupole-Orbitrap Mass Spectrometer (Thermo Fisher) coupled to a Vanquish UHPLC System (Thermo Fisher). Chromatographic separation was achieved on a Synchronis HILIC UHPLC column (2.1 mm × 100 mm × 1.7 µm; Thermo Fisher) using a binary solvent system (A + B) at a flow rate of 600 µL/min. Solvent A, 20 mM ammonium formate pH 3.0 in mass spectrometry grade H₂O; Solvent B, mass spectrometry grade acetonitrile (Thermo Fisher) with 0.1% formic acid (%v/v). A sample injection volume of 2 µL was used. The mass spectrometer was run in negative full-scan mode at a resolution of 240,000 scanning from 50 to 750 *m/z*. Following data curation and annotation, 86 metabolites remained in the dataset for experiment 1 and 122 for experiment 2.

Fecal lipid analysis

Fecal lipid concentration was determined from samples collected at 9 weeks of age. Mouse fecal pellets were weighed and homogenized in 500 µl of sterile PBS (Corning) at 15 Hz for 2 min. Total lipid content was isolated from 100 µl of homogenate with Lipid Extraction Kit (Chloroform free; Cell Biolabs Inc., USA) and lipid concentration was quantified in duplicate with Lipid Quantification Kit (Fluorometric; Cell Biolabs Inc.) according to manufacturer instructions. Results were quantified with the SpectraMax i3x (Molecular Devices, USA) and normalized to sample weight.

Oral glucose tolerance test

At 12 weeks of age, food was removed from cages at 0530-0600, followed by export from gnotobiotic isolators. Following 6 h of fasting, mice were gavaged with 50% dextrose (Sigma-Aldrich) at 2 g/kg body weight and blood glucose measurements were performed on tail blood at 15, 30, 60, 90 and 120 min with the OneTouch Ultra 2 glucometer (Lifescan, Switzerland). Body weight and fasted blood glucose were measured within 30 min of initiation of the test.

Adipose tissue histology and analysis

Peri-gonadal adipose tissue was dissected and placed in Dulbecco's Modified Eagle Medium without D-Glucose, L-Glutamine, Phenol Red or Sodium Pyruvate (DMEM; Gibco, USA) supplemented with 10% inactivated horse serum (HS; Sigma-Aldrich) and 12.5 mM HEPES (Gibco) for up to 1 h before a -1 cm piece of tissue was placed in 10% formalin. After 48 h, tissues were transferred to 70% ethanol for up to 2 weeks. Tissues were paraffin-embedded, sectioned at 4 µm and H&E stained by Alberta Precision Laboratories (APL). Images were acquired on the Aperio AT2 slide scanning microscope (Leica Biosystems, Germany) at 20X magnification. Adipocyte size was quantified using the Adiposoft plugin⁹⁰ in ImageJ with exclusion on edges. Three to four representative image sections were analyzed per sample.

Liver histology and analysis

Following wet weight measurement of the whole liver, the left lateral lobe was dissected and placed in 10% formalin. After 24 h, tissues were transferred to 70% ethanol for up to 2 weeks. Tissues were paraffin-embedded, sectioned at 4 µm and H&E stained by APL. Images were acquired on the Aperio AT2 slide scanning microscope (Leica Biosystems) at ×20 magnification. Lipid droplets were quantified in ImageJ with the Analyze Particles function with 0.9 circularity. Three to four representative image sections were analyzed per sample.

Plasma lipid, hormone and cytokine quantification

Following completion of the OGTT, animals were anesthetized with isoflurane and blood was drawn by cardiac puncture with a heparinized needle. Blood was combined with Dipeptidyl peptidase 4 inhibitor (Sigma-Aldrich) and Aprotinin (Sigma-Aldrich) and placed on ice for up to 2 h. Samples were spun at 2000 × *g* for 10 min and plasma was separated and stored at -80 °C until further processing. Plasma lipids were quantified by APL. Metabolic hormones and cytokines were quantified with U-PLEX Metabolic Group 1 Mouse Assay kit on the MESO QuickPlex SQ 120 (Meso Scale Discovery, USA). Plasma was diluted 4X in PBS and the assay was carried out following the manufacturer's instructions.

Preparation of fecal filtrates

Fecal samples were collected from mice at 6–13 weeks of age (B, B + C, B + R, B + M) and processed immediately. Approximately 200 mg of feces was weighed and diluted 1:5 with sterile PBS (Gibco). Samples were homogenized for 2 min at 15 Hz with the TissueLyser II bead beater (Qiagen) and centrifuged at 4000 × *g* at 4 °C for 5 min. The supernatant was filter sterilized with a 0.22 µm sterile syringe filter and stored at -80 °C for up to 1 week.

Enteroid culture and treatment

Murine jejunal enteroids were generated from intestinal crypts isolated from conventionally housed C57Bl/6J male mice at 20 weeks of age⁹¹, resuspended in Matrigel (Corning), and subsequently cultured in D-Mouse Complete Medium (50% WRN conditioned media and the other 50% comprised of advanced DMEM/F12 (Gibco) supplemented with GlutaMAX (2 mM), HEPES (10 mM), Penicillin-Streptomycin solution (Sigma), N2 supplement (1X), B27 supplement (1X), in addition to hEGF (50 ng/mL), N-Acetylcysteine (1 mM), Primocin (100 µg/mL; InvivoGen) + Y27632 (10 µM). Enteroids-derived monolayers were seeded between passages 7–12 at 1×10^6 cells per Matrigel-coated 3 µm transwell insert. Monolayers were grown in 2D-Mouse complete media (50% comprised of WRN conditioned media and the other 50% advanced DMEM/F12 (Gibco) supplemented with glutamax (200 mM), HEPES (1 M), Penicillin-Streptomycin solution (Sigma), N2 supplement (1X), B27 supplement (1X), in addition to hEGF (50 ng/mL), Primocin (100 µg/mL; InvivoGen) + Y27632 (10 µM) and media was changed the day after seeding and every other day after that. Once transepithelial electrical resistance reached 1000Ω, media containing 10% fecal filtrates was added to the transwell. 2D-Mouse complete media containing 20 ng/ml of IL-17 (R&D Systems) was added to the media outside of the transwell. Following 24 h of treatment, monolayers were collected in Trizol and stored at −80 °C.

RNA isolation and qPCR

RNA was isolated from 4 pooled monolayers with the RNeasy Mini kit (Qiagen). Reverse transcription was performed with the All-in-One 5X RT MasterMix (Applied Biological Materials Inc., Canada) and cDNA quality was assessed using the NanoDrop spectrophotometer (Thermo Fisher). PCR reactions were performed with iQ SYBR Green Supermix (BioRad) on the QuantStudio 3 Real-Time PCR System (Applied Biosystems). Twenty µL reactions were run in triplicate with the following cycling protocol: initial 2-min step at 94 °C, followed by 40 cycles of 94 °C for 15 s and 50 °C for 1 min and a final melt curve of 95 °C for 15 s, 60 °C for 1 min, 95 °C for 15 s and 60 °C for 15 s. Primers were as follows: *Actb* (forward, 5'-GTGACGTTGACATCCGTAAAGA-3'; reverse, 5'-GCCGGACTCATCGTACTCC-3'), *Cd36* (forward, 5'-ATGGGCTGTGATCGGAAGT-3'; reverse, 5'-GTCTTCCCAATAAGCATGTCTCC-3'), *Slc5a1* (forward, 5'-TG GTGTACGGATCAGGTCATTG-3'; reverse, 5'-TTCAGATAGCCACAGG GTACAG-3'). Fold gene expression was calculated with the $2^{-\Delta\Delta Ct}$ method with *Actb* serving as the housekeeping gene.

Adipose tissue immune cell isolation

Peri-gonadal adipose tissue was dissected and placed in DMEM (Gibco) supplemented with 10% HS (Sigma-Aldrich) and 12.5 mM HEPES (Gibco) on ice for up to 1 h. The wet weight of the tissue was obtained, and samples were mechanically minced and placed back in the media. One mg/mL of collagenase Type II (Sigma-Aldrich) and 5KU/mL of DNase I (Sigma-Aldrich) were added to the media and samples were incubated at 100 rpm 37 °C for 20 min with manual shaking every 5 min. Samples were supplemented with 0.1M EDTA and incubated at 37 °C with 10 rpm shaking for another 5–10 min. Once digestion was complete, samples were filtered through 100 µm mesh and topped up with 25 mL DMEM solution. Samples were spun at $300 \times g$ for 5 min at 4 °C. Supernatant was aspirated, and red blood cell lysis was performed with 2 mL ACK Lysing Buffer (Quality Biological) at room temperature for 2 or 7 min for samples from SD or HFHS mice, respectively. Samples were topped up with 8 mL cold PBS supplemented with 2% HS (Sigma-Aldrich), filtered through 250 µm mesh, and spun at $300 \times g$ for 5 min at 4 °C. Supernatant was aspirated, and the cell pellet was resuspended in RPMI Medium 1640 (Gibco) supplemented with 10% HS.

Splenocytes isolation

Spleens were dissected and placed in RPMI (Gibco) supplemented with 5% HS (Sigma-Aldrich) on ice for up to 2 h. The wet weight of the tissue

was obtained, and samples were mechanically minced. One mg/mL of Collagenase Type IA (Sigma-Aldrich) and 30 U/mL of DNase I were added to the media and samples were incubated at 220 rpm 37 °C for 25 min with mechanical shaking halfway through. Samples were filtered through a 100 µm cell strainer, topped up with 5 mL PBS supplemented with 2% HS and spun at $400 \times g$, 5 min 4 °C. Supernatant was removed and red blood cell lysis was performed with 5 mL ACK Lysing Buffer (Quality Biological) at room temperature for 7 min. Samples were topped up with 15 mL cold PBS supplemented with 2% HS (Sigma-Aldrich) and spun at $400 \times g$ for 5 min at 4 °C. Supernatant was removed, and the pellet was resuspended in RPMI (Gibco) supplemented with 10% HS.

Immunolabelling and flow cytometry

Isolated immune cells from adipose tissue and splenocytes were stained with Fixable Viability Stain (eF506; eBioscience) and stained for extracellular immune cell markers. Cells were then fixed and permeabilized with the Foxp3/Transcription Factor Staining Buffer Set (eBioscience, USA) and stained for intracellular immune cell markers. Samples were stained with a lymphoid and myeloid panel. The following antibodies were included in the lymphoid panel: anti-CD45 AF532 (clone 30-F11; eBioscience), anti-TCRb BV750 (clone H57-597; BD), anti-γδTCR PE-CF594 (clone GL3; BD), anti-CD4 BV786 (clone RM4-5; BD), anti-CD8 BV570 (clone 53-6.7; BioLegend), anti-CD19 PerCP-Cy5.5 (clone 1D3; BD), anti-NK1.1 BUV395 (clone PK136; BD), anti-CD90.2 BUV496 (clone 30-H12; BD), anti-T-bet BV421 (clone 4B10; BioLegend), anti-GATA3 BV711 (clone L50-823; BD), anti-FOXP3 AF488 (clone FJK-16s; eBioscience), anti-RORγt PE (clone B2D; eBioscience), anti-Eomes PE-Cy5 (clone Dan11mag; eBioscience), anti-IL-5 eF450 (clone TRFK5; eBioscience), anti-IL-10 BV605 (clone JES5-16E3; BD), anti-IFN-γ BV650 (clone XMGL2; BD), anti-IL-17A PE-Cy7 (clone eBio17B7; eBioscience). The following antibodies were included in the myeloid panel: anti-CD45 AF532 (clone 30-F11; eBioscience), anti-CD11b BUV395 (clone M1/70; BD), anti-SigF BV786 (clone E50-2440; BD), anti-Ly6G BV750 (clone 1A8; BD), anti-CD64 BV650 (clone X54-5/7.1; BD), anti-CD11c AF647 (clone N4/18; BioLegend), anti-MHCII eF450 (clone M5/114.15.2; eBioscience), anti-CD103 PE-Dazzle594 (clone M290; BioLegend), anti-Ly6C BV570 (clone HK1.4; BioLegend), anti-CX3CR1 BV421 (clone SA011F11; BioLegend), anti-Tim4 PE-Cy7 (clone RMT4-54; BioLegend), anti-CD206 AF700 (clone C068C2; BioLegend), anti-CD9 FITC (clone MZ3; BioLegend). Samples were run on the Cytex Aurora (Cytex Biosciences, USA). The same gating strategy was used for both tissue types, except for macrophage populations, due to low CD64⁺ population in spleens and our interest in WAT-specific macrophage populations.

Statistics and reproducibility

Sample size calculations were not performed. We aimed to achieve $n = 8$ –10 animals/group based on the literature for DIO studies in mice^{92–94}, though some limitations in this were due the distribution of mouse sex among the litters. Mouse litters were assigned to the different diet groups in an attempt to evenly distribute litter sizes. This was dependent on breeding rates and birth. Mice were housed in different isolators based on colonization status (i.e., B, B + C, B + R, B + M). The Investigators were not blinded to allocation during experiments and outcome assessment as there was no subjective component. Data was excluded from certain analyses if they were identified as outliers by ROUT ($Q = 1\%$). This is specified in the figure captions. Given the low expression levels of *Cd36* and *Slc5a1* in jejunal enteroids, samples were excluded from qPCR analysis if triplicates differed by greater than 2 cycles⁹⁵.

Bacterial alpha-diversity was assessed with the Shannon diversity index using *vegan* v2.6-4⁹⁶. Data was assessed for normality with the Shapiro–Wilk test, and differences between fungal colonized groups and their respective “B” control groups (sequenced in the same run)

were assessed by unpaired *t*-test (parametric) or Mann–Whitney (non-parametric) test accordingly. Boxplots display the median and hinges, with the lower and upper hinges displaying the 25th and 75th percentile, respectively. Whiskers extend from the hinge to the lowest/largest value below 1.5X the inter-quartile range. Beta-diversity was assessed with the Bray–Curtis dissimilarity index with variance-stabilizing transformation and visualized with principal coordinate analysis (PCoA), where ellipses represent the 95% confidence interval. Differences between fungal colonized groups and their respective B-only group, along with sex differences, were assessed by permutational ANOVA at 3 weeks, with the addition of a diet effect at 12 weeks using *vegan* v2.6-4⁹⁶. The relative abundance of the 12 bacterial strains was compared between fungal colonized groups and their respective B-only group by unpaired *t*-test or Mann–Whitney test depending on the normality of the data, as assessed by the Shapiro–Wilk test.

For metabolic data collected at a single time point, data was first assessed by Three-way ANOVA to determine the presence of sex effect. If present, males and females were analyzed separately; otherwise, males and females were pooled for analysis. Outliers were identified with the ROUT method ($Q = 1\%$) and excluded as indicated in the figure legends. The effect of diet and colonization was then analyzed by Two-way ANOVA, where fungal colonized groups were compared to B-only groups pooled from all experiments within each diet with Dunnett's multiple comparison test. Longitudinal measures (i.e., body weight, OGTT), were first analyzed by RM-ANOVA (MANOVA.RM v0.5.4⁹⁷), where colonization, diet and sex were included as the between subject factors and time was the within subject factor. If the effect of sex was significant, males and females were analyzed separately. Individual time points in either sex for body weight, during the OGTT and trapezoidal AUC were compared using Two-way ANOVA with Dunnett's multiple comparison test. Immune data for either sex on either diet were assessed for normality by Shapiro–Wilk test, and fungal colonized groups were compared to B-only groups by One-way ANOVA with Dunnett's multiple comparison test or Kruskal–Wallis test with Dunn's multiple comparison test, accordingly. Bar and line plots display the mean with standard error of the mean (SEM).

Metabolomics data was processed with Metaboanalyst v5.0⁹⁸. Metabolite concentrations were normalized by the median, square root transformed and subject to Pareto scaling. Euclidean distance calculations from PCA scores and permutational ANOVA were performed using *vegan* v2.6-4⁹⁶. Fungal groups were compared to the “B” group from the same experiment that was processed in the same batch. Direction of comparison for fold change analysis was specified as fungal groups/B-only and test parameters were defined as non-parametric with unequal variance and false discovery rate ($p < 0.05$) correction. *MIMOSA2* v2.0.0⁴⁴ was performed using a customized AGORA database that included only the Oligo-MM12 strains and the same ASVs identified with *DADA2* were mapped to the database with a similarity threshold of 0.99. Metabolite data was log-transformed, and the regression model was fit using rank-based estimation with the significance threshold set to 0.05.

For Spearman correlation analysis, study variables were grouped as colonization, metabolism, bacterial abundance or WAT immunity. Data underwent mean normalization and correlations between variables from the different groups were identified using *Hmisc* v5.1.1⁹⁹ package. Significant correlations (coefficient $> |0.6|$) were plotted with *circlize* v0.4.15¹⁰⁰.

Flow cytometry data was gated using FlowJo (v10.8.1). Graphs were made using either RStudio (v2022.12.0 + 353) or GraphPad Prism v9.5.1.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The demultiplexed 16S rRNA amplicon sequencing data generated in this study have been deposited in the Sequence Read Archive (SRA) of NCBI under accession code BioProject [PRJNA1041856](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1041856). Metabolomics mass spectral raw data can be shared upon request. The metabolic and immune data generated in this study are provided in the Supplementary Information/Source Data file. All microbial strains are kept as stocks in the Arrieta lab and can be shared when requested. Source data are provided with this paper.

Code availability

Code can be found at https://github.com/ArrietaLab/Mycobiome_Metabolic_Health_Mice (<https://doi.org/10.5281/zenodo.14611152>)¹⁰¹.

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Conceptualization: M.W.G. and M.-C.A. Methodology: M.W.G., F.C., S.A.H., R.A.R., and M.-C.A. Formal analysis: M.W.G. and E.L.L. Investigation: M.W.G., E.V.T.B., E.R., K.N.K., M.D., E.L.L., T.G., E.M.M., H.Z., S.S., and A.A. Writing—original draft: M.W.G. and M.-C.A. Writing—reviewing and editing: M.W.G., E.V.T.B., K.N.K., E.L.L., T.G., H.Z., E.M.M., S.S., F.C., S.A.H., R.A.R., and M.-C.A. Funding acquisition: M.-C.A. Resources: M.-C.A. Supervision: M.-C.A.

Competing interests

The authors declare no competing interests.

Additional information

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